

Data Mining Questions

Clustering / K-Means Clustering

1. What is clustering? Write short note on K-means algorithm.
2. Suppose that the data mining task is to cluster points (with (x, y) represent location) into three clusters, where points are $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, $C2(4, 9)$. The distance function is the Euclidean distance / Manhattan distance / Minkowski distance. Suppose initially we assign $A1$, $B1$, and $C1$ as the center of each cluster, respectively. Use the k-means algorithm to show only
 - a. The three cluster centers after the first round of execution
 - b. The final three clusters
3. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8)
 - a. Compute the Manhattan distance between two objects
 - b. Compute the Minkowski distance between the two objects, using $q=4$.

Classification / SVM

1. Define kernel trick. Discuss about different kernel Support Vector Machines.
2. In Support Vector Machines, what is a support vector? Write short notes on SVM?
3. Positively labelled data points (3, 1), (3, -1), (6, 1), (6, -1) and negatively labelled data points (1, 0), (0, 1), (0, -1), (-1, 0). Find the Hyper-plane coordinates with bias and weight.
4. Positively labelled data points (2, 2), (2, -2), (-2, -2), (-2, 2) and negatively labelled data points (1, 1), (1, -1), (-1, -1), (-1, 1). Find the Hyper-plane coordinates with bias and weight. Where kernel SVM condition:

$$\theta_1 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{cases} \begin{pmatrix} 4 - x_2 + |x_1 - x_2| \\ 4 - x_1 + |x_1 - x_2| \end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} > 2 \\ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} & \text{otherwise} \end{cases}$$

Dimensionality Reduction / PCA

1. What is understood by curse of dimensionality? How can we avoid this problem?
2. Briefly discuss the steps of Principle Component Analysis (PCA) technique.
3. Given data points {2, 3, 4, 5, 6, 7; 1, 5, 3, 6, 7, 8}
 - a. Compute the principle component using PCA Algorithm.

- b. Use PCA algorithm to transform the pattern (2, 1) onto the Eigen vector in the previous question

Association rule mining / Apriori Algorithm

1. What are association rule in data mining. Give the names of algorithm in association rule.
2. Write short notes on Apriori algorithm. What are the components of Apriori algorithm?
3. From the table below, find frequent item-sets and generate the association rule using the Apriori algorithm: (*maximum support = 2, minimum confidence = 50%*)

TID	ITEM SETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

Introduction / Common

1. In context of machine learning, define training data, testing data, data mining, supervised, unsupervised and reinforcement learning
2. Define feature, feature extraction and data processing with example
3. What are the primary differences between machine learning, data mining, and pattern recognition?
4. From the following confusion matrix, find out –

N = 165	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100

Accuracy, Precision, Recall, F1 – score, FN rate, TN rate

5. What are outliers? How might you determine outliers in the data?
6. What is the basic difference between supervised and unsupervised learning?
7. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
 - a. What is the mean of the data? What is the median?
 - b. What is the mode of the data? Comment on the data's modality
 - c. What is the midrange of the data?